

Michael C. Eaton

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Education

Data Science | Masters of Science

Aug '25 - Dec '26

Rochester Institute of Technology | ROCHESTER, NY

- GPA: 4.0
- **Relevant Coursework:** Applied Statistics, Machine Learning, Simulation Optimization, Database Systems, Digital Twins

Game Design and Development | Bachelors of Science

Aug '22 - Dec '24

Rochester Institute of Technology | ROCHESTER, NY

- **Awards:** Dean's List (All Semesters 2022–2024)

Work experience

Research Assistant - Agentic AI Sandbox

Jun '26 - Present

Rochester Institute of Technology | ROCHESTER, NY

- Built a 9-agent classification pipeline enforcing a hard trust boundary: agents propose, a deterministic harness re-validates, sandboxes, and gates every candidate before promotion
- Designed a dynamic planning agent that adaptively sequences the pipeline by goal, validated via parity testing and an adversarial prompt-injection evaluation
- Generalized the pipeline to multiclass targets and a 4.2GB aviation time-series dataset, adding an explainability agent that localizes flagged predictions via occlusion attribution

Research Assistant - Lighting Estimation, DISCO Research Lab

May '26 - Present

Rochester Institute of Technology | ROCHESTER, NY

- Built a diffusion-based model (fused encoder, render + perceptual VGG loss) to predict per-scene lighting “basis images” from a single photo
- Developed synthetic data pipelines (Blender, variable camera distance) and validated accuracy via NNLS energy-recovery experiments and grid-density sweeps

Teaching Assistant - Rich Media Web Development

Jan '26 - Present

Rochester Institute of Technology | ROCHESTER, NY

- Provide technical mentorship for undergraduate students, utilizing advanced debugging techniques to resolve complex logic errors and unblock student progress during development cycles
- Enforce industry-standard engineering practices during code reviews, emphasizing proper Git workflows, modular architecture, and clean code principles to prepare students for professional roles

AI Trainer - Coding Expert

Oct '25 - Apr '26

Confidential | REMOTE

- Conducted line-by-line technical QA on 200+ Python and Java code snippets, debugging syntax errors and edge cases to ensure training data functionality
- Evaluated model reasoning across 5+ turn conversations, ranking outputs based on logical consistency, hallucination rate, and adherence to complex prompt constraints
- Maintained a 99% accuracy rating on technical evaluations, exceeding the project's quality threshold

Development Team Lead - Peaceland

Aug '24 - Dec '24

Rochester Institute of Technology | ROCHESTER, NY

- Led a cross-functional team of 20 developers and designers, implementing Agile workflows to coordinate feature integration and deliver a playable Unity prototype on a 4-month timeline

Projects

Diabetic Patient Readmission Modeling

Oct '25 - Dec '25

Rochester Institute of Technology | ROCHESTER, NY

- Designed an end-to-end Machine Learning pipeline to predict hospital readmission, handling data imbalance and feature engineering to improve patient care
- Conducted a cost-benefit analysis using a financial cost matrix to identify an optimal decision threshold, projecting an average saving of \$37.69 per patient (approx. \$37k per 1,000 admissions)

Deep Reinforcement Learning for Dynamic Flexible Job Shop Scheduling

Dec '25 - Present

- Building a Deep Reinforcement Learning (DRL) framework to optimize production scheduling in spatially constrained environments
- Developing a high-fidelity Unity3D simulation environment to address the Dynamic Flexible Job Shop Scheduling Problem (DFJSP)
- Designing a benchmarking pipeline to stress-test agent adaptability against randomized failure scenarios

Skills

- **AI & Machine Learning:** Python (Scikit-learn, Pandas, NumPy, PyTorch, TensorFlow), Deep Reinforcement Learning (DRL), LLM Evaluation/RLHF, Context Analysis
- **Simulation & Optimization:** Linear Programming (LP), Stochastic Modeling, Digital Twins, Unity3D (C#)
- **Data & Tools:** SQL (PostgreSQL), AWS, Git/Version Control, API Integration (REST)